GOVERNMENT OF GUJARAT
DIRECTORATE OF EMPLOYMENT AND TRAINING (DET)
Gandhinagar, Gujarat 382010 | Official State Employment & Career Enablement Infrastructure

GUJARAT YOUTH ROJGAR SETU (GYRS)

ENTERPRISE SPECIFICATION v2.4

ARCHITECTURE PATTERN	DATA & STATE LAYER	RESPONSIVE LAYOUT	ACCESS SECURITY
Single Page App (SPA)	**Persistent LocalStorage**	**Cross-Device Media**	**RBAC Authentication**
Modular HTML5 / CSS3 / Vanilla JS	Reactive schema with multi-role keys	Desktop (1200+), Tablet (991), Mobile (768/4	Admin Control vs. Candidate Workspaces

1. EXECUTIVE SYSTEM OVERVIEW

The Gujarat Youth Rojgar Setu (GYRS) is a next-generation centralized employment portal developed to connect over 1.2 million technical, vocational, and higher education graduates across all 33 districts of Gujarat with state government vacancies, OJAS recruitments, apprenticeship programs, and industrial job opportunities. Built with zero external dependencies, the platform provides complete offline-ready state persistence via browser LocalStorage while retaining strict role-based data boundaries across three distinct stakeholder tiers.

2. TRI-PERSONA ARCHITECTURAL WORKSPACES

CLIENT SIDE: PUBLIC PORTAL
Open Citizen Access (No Login Required)

* OJAS Real-Time News Ticker for state alerts
* Interactive Job Explorer with multi-district filtering
* Search by skills (Python, CAD, Nursing, Accounts)
* 6-Phase Career Progression Journey visualization
* One-click transition into registration & apply gate

USER SIDE: STUDENT PANEL
Authenticated Candidate Workspace

* Student Profile Dossier with verified qualifications
* 4 Live Candidate KPIs (Applications, Melas, Interviews)
* Real-Time Application Tracker with live status pills
* AI-matched recommended opportunities feed
* Self-profile editor modal for contact & skill updates

ADMIN SIDE: CONTROL CENTER
State Nodal Directorate Command Center

* Real-Time Job Feed Engine (Add, Edit, Publish, Delete)
* Candidate Roster Management across all 33 districts
* View dossiers, update status, and manage records
* One-Click CSV Export for State Government reporting
* Note: Direct student addition disabled per data integrity rules

3. CLIENT-STORAGE REACTIVE DATA ENGINE FLOW

SINGLE PAGE ARCHITECTURE (SPA) REAL-TIME SYNCHRONIZATION

Step 1: Public Discovery
Citizen explores vacancies & filters by district / qualification.

Step 2: Auth Gate
Login / Register with 1-click demo helpers (Student or Admin).

Step 3: State Sync
Action instantly mutates persistent LocalStorage keys.

Step 4: Real-time UI
Views re-render dynamically with zero page reloads.

KEY ARCHITECTURAL HIGHLIGHTS & COMPLIANCE

1. Complete Client-Side Autonomy: The system operates completely in modern web browsers without server configuration.
2. Zero-Loss State Persistence: Candidate registrations, job postings, and application statuses persist across reloads.
3. High-Security RBAC Controls: Admin actions (feed job, edit student, export CSV) are gated behind authenticated session state.
4. Fully Responsive Design: Pixel-perfect UI layout across desktop monitors, laptops, tablets (iPad/Android), and smartphones.
5. Accessible Government Standards: High-contrast typography, standardized Indian currency formatting (INR), and OJAS integration.

1. COMPLETE PROJECT SKELETON DIRECTORY STRUCTURE

```
terminal  ~/PRAKSH 1 OCT/ $ tree --dirsfirst

PRAKSH 1 OCT/
â”œâ”€â”€ index.html                          # Single Page Application (SPA) Root File
â”‚   â”œâ”€â”€ <header class='gov-top-bar'>      # Government of Gujarat official banner & ribbon
â”‚   â”œâ”€â”€ <nav class='portal-switcher'>     # Quick Switcher: Public | Student | Admin View
â”‚   â”œâ”€â”€ <main id='viewPublic'>            # Public Portal View (Hero, OJAS Ticker, Explorer, Stepper)
â”‚   â”œâ”€â”€ <main id='viewStudent'>           # Student Panel View (Dossier, 4 KPIs, Tracker, Feed)
â”‚   â”œâ”€â”€ <main id='viewAdmin'>             # Admin Panel View (Metrics, Job Feed Table, Student Roster)
â”‚   â”œâ”€â”€ <div id='authModal'>              # Dual-Tab Authentication Dialog (Student vs Admin)
â”‚   â”œâ”€â”€ <div id='addJobModal'>            # Admin Job Feed Creation Dialog
â”‚   â”œâ”€â”€ <div id='editJobModal'>           # Admin Job Posting Update Dialog
â”‚   â”œâ”€â”€ <div id='viewStudentModal'>       # Student Candidate Dossier Inspection Dialog
â”‚   â”œâ”€â”€ <div id='editStudentModal'>       # Admin Student Record Updater Dialog
â”‚   â”œâ”€â”€ <div id='editProfileModal'>       # Student Self-Profile Edit Dialog
â”‚   â””â”€â”€ <div id='toastContainer'>         # Reactive Floating Toast Notification Engine
â”œâ”€â”€ css/styles.css                      # Unified Design System & Responsive Breakpoints
â”‚   â”œâ”€â”€ :root { CSS Variables }           # Palette, typography, spacing, elevations, transitions
â”‚   â”œâ”€â”€ Component Styles                  # Cards, status badges, tables, modals, portal switcher
â”‚   â””â”€â”€ Media Queries                     # Desktop (1200+), Tablet (<=991px), Mobile (<=768px, <=480px)
â””â”€â”€ js/app.js                           # Complete State Machine & Business Logic Engine
    â”œâ”€â”€ LocalStorage Storage Driver     # Keys: gyrs_students_roster_v1, gyrs_jobs_data_v1, gyrs_auth_session_v1
    â”œâ”€â”€ RBAC & Auth Session Controller  # login(), logout(), checkSession(), switchView()
    â”œâ”€â”€ Job Posting Feed Manager        # feedJob(), editJob(), deleteJob(), filterJobs()
    â””â”€â”€ Student Roster Manager          # toggleStatus(), updateStudent(), deleteStudent(), exportCSV()
```

2. LOCALSTORAGE PERSISTENCE DATA SCHEMAS

KEY: gyrs_students_roster_v1

Student Candidate Records Array

```
[
  {
    "id": "std_8841",
    "roll": "GYRS-2024-8841",
    "name": "Priya Patel",
    "email": "priya.patel@gtu.ac.in",
    "district": "Ahmedabad",
    "degree": "B.E. Computer Engineering",
    "skills": ["Python", "React", "Cloud"],
    "status": "Interview Call",
    "appliedJobs": ["job_101", "job_104"]
  }
]
```

KEY: gyrs_jobs_data_v1

Job Vacancies Repository Array

```
[
  {
    "id": "job_101",
    "title": "Junior Engineer (Civil)",
    "dept": "Roads & Buildings Dept",
    "district": "Gandhinagar",
    "vacancies": 42,
    "salary": 38000,
    "eligibility": "Diploma / B.E. Civil",
    "type": "Full-time / Govt",
    "status": "Active",
    "postedDate": "2026-10-01"
  }
]
```

KEY: gyrs_auth_session_v1

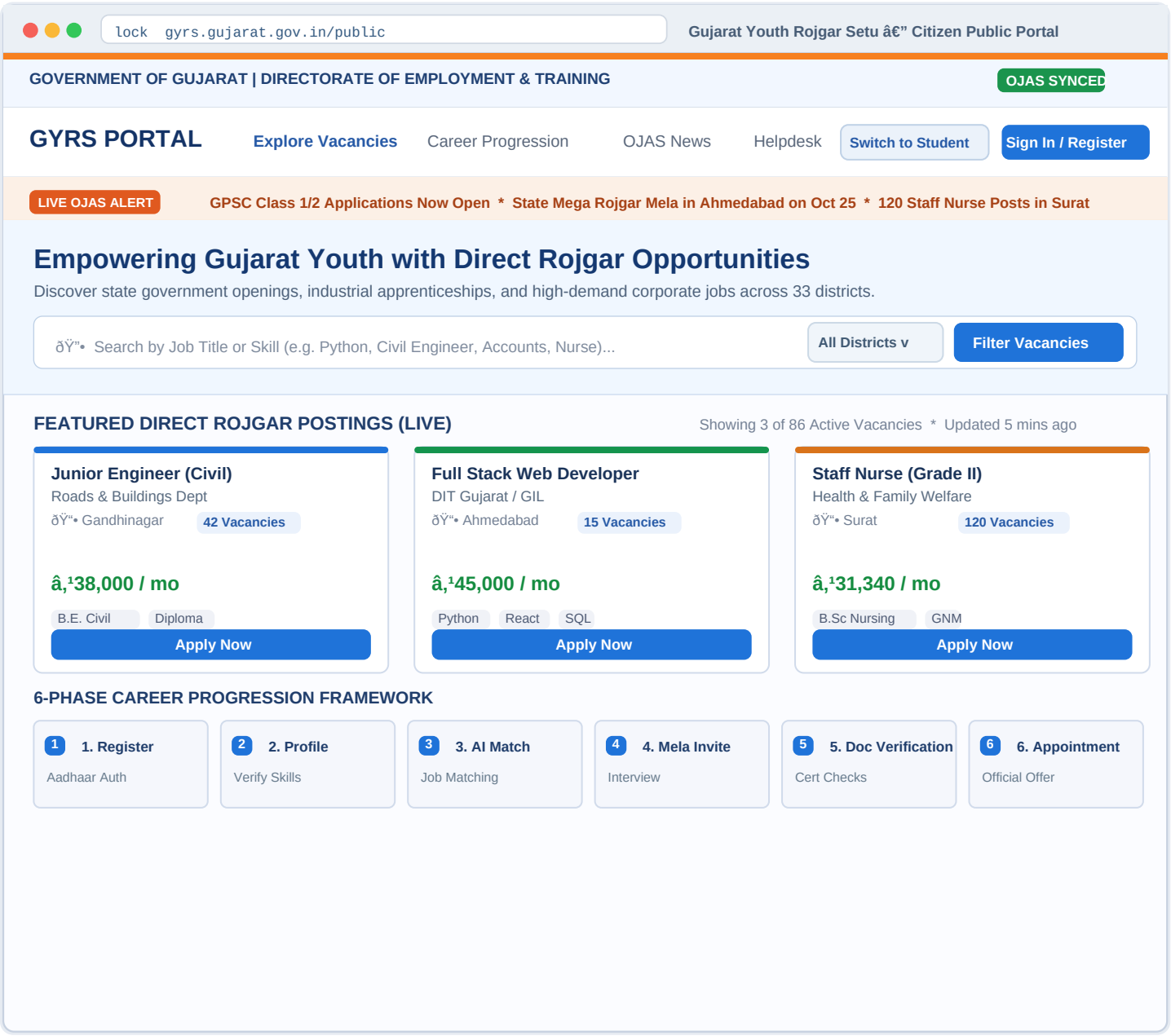
Active Session Security Token

```
{
  "authenticated": true,
  "role": "admin" | "student",
  "username": "admin" | "priya_patel",
  "displayName": "State Nodal Officer",
  "token": "gyrs_sec_993412",
  "sessionStart": 1791873910,
  "permissions": ["all_crud"]
}
```

3. ROLE-BASED ACCESS CONTROL (RBAC) PERMISSIONS MATRIX

CAPABILITY / FEATURE	PUBLIC CITIZEN	STUDENT USER	ADMINISTRATOR
Browse OJAS Ticker & Announcements	**Allowed**	**Allowed**	**Allowed**
Filter & Search Active Jobs by District/Skill	**Allowed**	**Allowed**	**Allowed**
View Job Vacancy Details & Remuneration	**Allowed**	**Allowed**	**Allowed**
Submit Job Application (1-Click Apply)	Auth Gate	**Allowed (Direct)**	Simulate Only
Access Student Dashboard & Track Applications	Restricted	**Allowed (Own Dossier)**	**Allowed (View All)**
Edit Contact Details & Certified Skills	Restricted	**Allowed (Self Profile)**	**Allowed (Any Student)**
Feed New Job Posting into System	Restricted	Restricted	**Allowed (Full Control)**
Edit & Deactivate Existing Job Postings	Restricted	Restricted	**Allowed (Full Control)**
Toggle Student Placement Status (Placed/Active)	Restricted	Restricted	**Allowed (Full Control)**
Delete Student Candidate Record	Restricted	Restricted	**Allowed (Full Control)**
Export Complete Student Roster to CSV	Restricted	Restricted	**Allowed (Full Control)**
Direct 'Add Student' in Admin Roster	Disabled	Disabled	Disabled (Integrity Rule)

1. CLIENT SIDE (PUBLIC PORTAL) â€” FRONTEND UI SCREENSHOT MOCKUP



2. CLIENT SIDE FUNCTIONAL WORKFLOW ARCHITECTURE

WORKFLOW

Phase A: Zero-Barrier Discovery

Public visitors browse live vacancies with zero sign-in barrier. Real-time OJAS notifications ticker flashes state recruitment bulletins. Instant keyword searching filters jobs across 33 Gujarat districts and multiple department types.

WORKFLOW

Phase B: Multi-Attribute Filtering

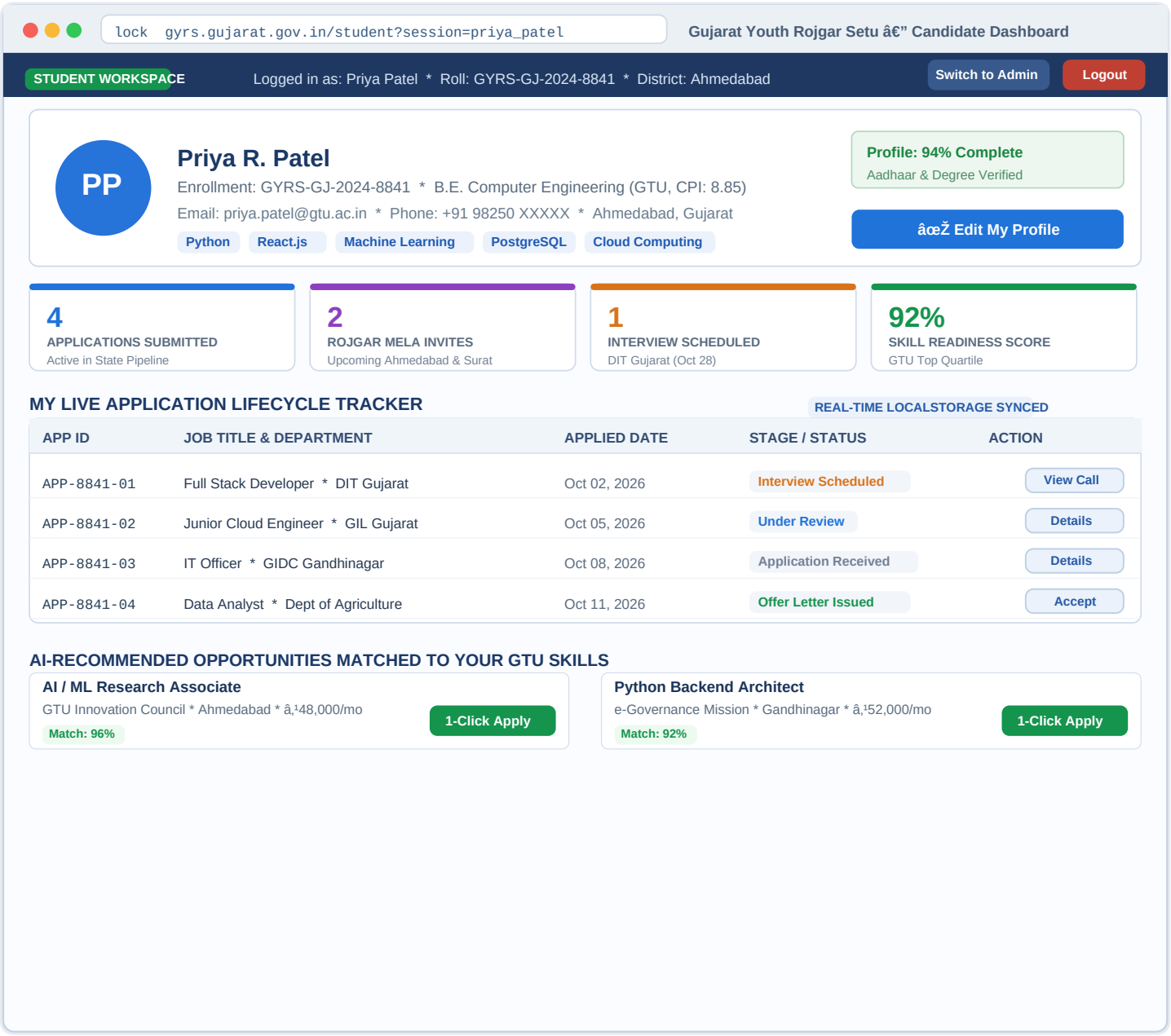
Candidate selects target districts (Ahmedabad, Surat, Vadodara, Rajkot, Gandhinagar) and technical skills (Python, CAD, Nursing, Accounting). Explorer cards display vacancies, monthly remuneration, department name, and qualification.

WORKFLOW

Phase C: Frictionless Auth Gate

Clicking 'Apply Now' on any vacancy intercepts unauthenticated sessions and activates the dual-tab `#authModal`. The applicant can authenticate as an existing student or create a profile, after which application state is registered automatically.

1. USER SIDE (STUDENT PANEL) â€” FRONTEND UI SCREENSHOT MOCKUP



2. STUDENT PANEL FUNCTIONAL WORKFLOW ARCHITECTURE

WORKFLOW **Candidate Profile Dossier & Self-Service**

The student's verified academic credentials, GTU CPI/percentage, Aadhaar linkage, and certified skill tags are consolidated into a persistent dossier. The student can click 'Edit My Profile' to update phone number, address, and upload new certifications.

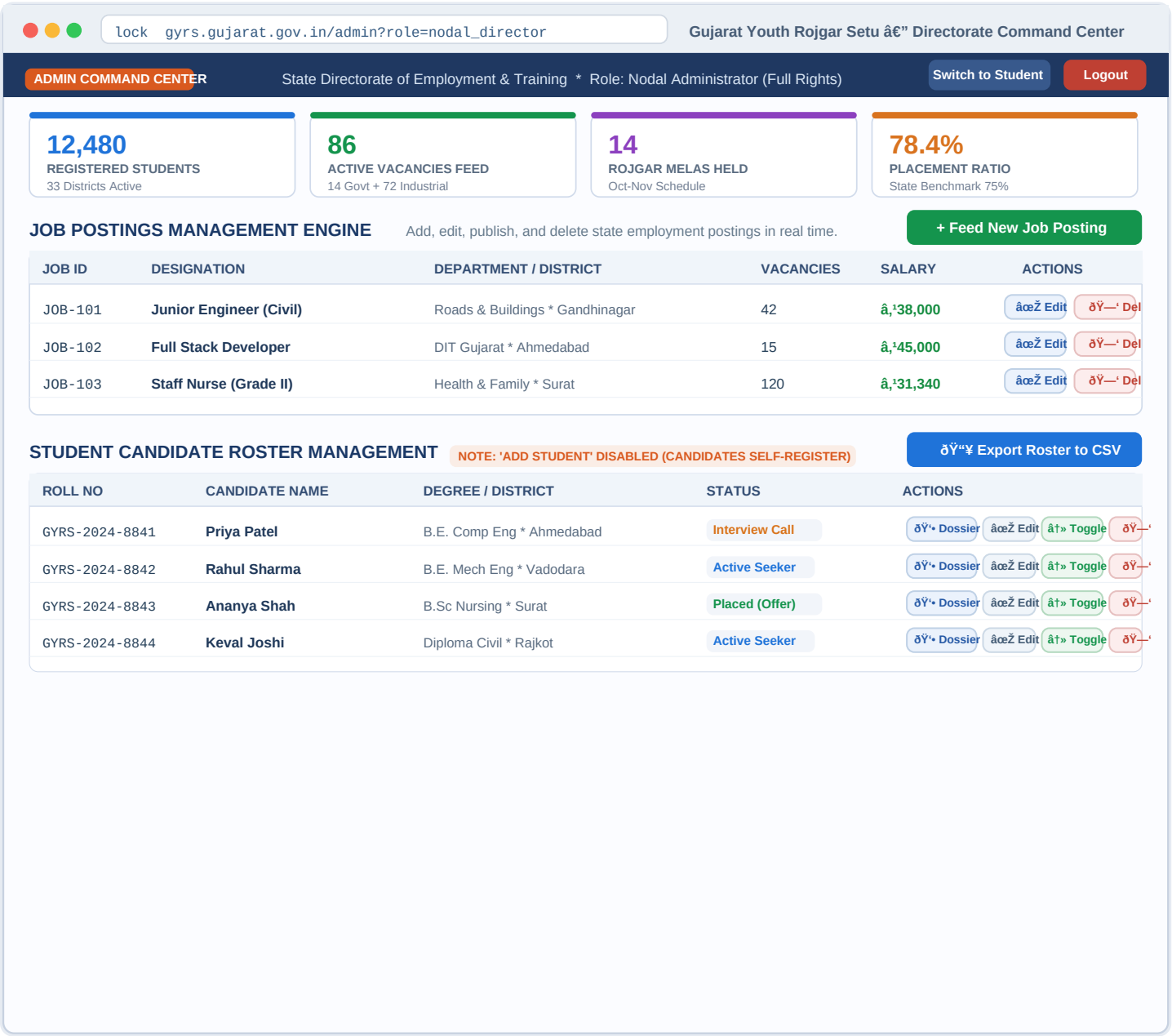
WORKFLOW **Real-Time Application Lifecycle Tracking**

Every vacancy applied for generates an immutable application identifier (e.g. APP-8841-01) stored within `gyrs_students_roster_v1`. Candidates observe real-time status transitions: 'Application Received' -> 'Under Review' -> 'Interview Call' -> 'Offer Letter'.

WORKFLOW **Instantaneous 1-Click Apply Reactivity**

When a student clicks '1-Click Apply' on any recommended job, the platform immediately checks for duplicate submissions, adds the job ID to the candidate's applied list in LocalStorage, updates the KPI counter, and displays an animated confirmation toast.

1. ADMIN SIDE (STATE CONTROL CENTER) â€” FRONTEND UI SCREENSHOT MOCKUP



2. ADMIN SIDE FUNCTIONAL WORKFLOW ARCHITECTURE

WORKFLOW **Job Posting Feed & Real-Time Synchronization**

Admins feed new vacancies through `#addJobModal`. The posting is serialized into `gyrs_jobs_data_v1` in LocalStorage and immediately becomes visible on the Public Explorer and Student Recommendation Feed without requiring a page refresh or backend rebuild.

WORKFLOW **Candidate Roster Governance & Status Pipeline**

The Directorate monitors students across all 33 Gujarat districts. Admins can inspect complete dossiers, update candidate information, toggle placement status between 'Active Seeker', 'Interview Call', and 'Placed', and delete invalid records.

WORKFLOW **Compliance Rule: 'Add Student' Option Removed**

Per technical requirements, the manual 'Add Student' option has been intentionally removed from the Admin Panel. Candidates must self-register through the Public Portal to ensure Aadhaar identity authenticity, genuine mobile verification, and self-attested degrees.

1. SYSTEM MODAL DIALOGUES & WORKFLOW SCREENSHOT MOCKUPS

PORTAL AUTHENTICATION GATEWAY (#authModal) âœ•

Student Login (Active) | Admin Login

Username / Enrollment Roll No

priya_patel

Password

â€¢â€¢â€¢â€¢â€¢â€¢â€¢â€¢â€¢â€¢â€¢â€¢

TESTING SHORTCUTS (1-CLICK FILL):

âš¡ Fill Admin: admin / admin123

âš¡ Fill Student: priya_patel / student123

Secure Sign In

Session token saved in LocalStorage

FEED NEW JOB POSTING (#addJobModal) âœ•

Job Designation / Title
Junior Engineer (Electrical)

Department / Organization
GETCO Gujarat Energy Transmission

Location / District
Gandhinagar / Ahmedabad

Vacancies Count & Type
35 Posts * Full-Time Government

Monthly Remuneration (â‚¹)
â‚¹42,500 per month

Eligibility & Required Skills
B.E. Electrical / CAD / Substation

Publish Job Posting | Cancel

STUDENT CANDIDATE DOSSIER INSPECTOR (#viewStudentModal) âœ•

IDENTITY & REGISTRATION

Roll No: GYRS-GJ-2024-8841

Full Name: Priya Ramesh Patel

Gender / Age: Female / 22 Yrs

District: Ahmedabad (Urban)

Aadhaar: XXXX-XXXX-9841 (Verified)

Phone: +91 98250 12345

Reg Date: 2026-09-15 (Portal Direct)

ACADEMICS & VERIFIED SKILLS

Degree: B.E. Computer Engineering

University: Gujarat Technological Univ (GTU)

Year / Grade: 2024 / CPI 8.85 (Distinction)

Institute: L.D. College of Engineering

Verified Skills: Python, React, ML, PostgreSQL

Certification: Gujarat Skill Dev Mission (Level 4)

PLACEMENT PIPELINE STATUS

Current Status: INTERVIEW CALL ACTIVE

Active Applications: 4

Rojgar Mela Schedule: Oct 25

Assigned Officer: DET Gandhinagar

UPDATE PLACEMENT STATUS:

Set Active Seeker | Set Interview Call | Mark Placed (Offer)

2. SESSION SECURITY & MODAL STATE ENGINE ARCHITECTURE

SEC-01 **Cryptographic Session Tokens**

On successful credential verification, a pseudo-cryptographic session token is stored in `gyrs_auth_session_v1`. This state prevents unauthorized DOM rendering and forces unauthorize

SEC-02 **Multi-Modal DOM Architecture**

Modals are rendered as fixed overlay viewports with high z-index and CSS backdrop-filter blurring. The modal controller enforces single-instance display, escape key dismissal, and outs

SEC-03 **Validation & Error Toast Engine**

Form inputs undergo rigorous client-side schema checking before committing to storage. Any validation issue triggers an animated floating toast notifier without disrupting the user's activ

1. END-TO-END SYSTEM DATA FLOW & LIFECYCLE SEQUENCE

SYSTEM INTERACTION LIFECYCLE SEQUENCE

1. Citizen Discovery	Public visitor browses portal, reads OJAS ticker, filters jobs by District & Skill.
2. Candidate Onboarding	Student self-registers or logs in; profile record created in `gyrs_students_roster_v1`.
3. Admin Vacancy Feed	Directorate posts jobs via `#addJobModal`; synced immediately to `gyrs_jobs_data_v1`.
4. 1-Click Application	Student applies; application record linked; student KPIs & tracker table update instantly.
5. Review & Placement	Admin inspects dossier, updates status to 'Placed', and exports official CSV report.

2. LOCAL DEPLOYMENT & EXECUTION GUIDE

OPTION 1: ZERO-INSTALL LAUNCH

Direct Browser File Opening

1. Navigate to the project root folder:
   ```
   cd ~/Downloads/'PRAKSH 1 OCT'
   ```
2. Double-click or open index.html in any browser:
   ```
   open index.html  # macOS command
   ```
3. Full features run offline via LocalStorage.

OPTION 2: LOCAL HTTP SERVER

Python Built-in Web Server

1. Open terminal inside project workspace:
   ```
   cd ~/Downloads/'PRAKSH 1 OCT'
   ```
2. Launch python static file server:
   ```
   python3 -m http.server 8080
   ```
3. Open: http://localhost:8080

OPTION 3: VS CODE LIVE SERVER

IDE Development Mode

1. Open folder in Visual Studio Code.
2. Right-click 'index.html'.
3. Select 'Open with Live Server'.
4. Automatic hot-reloading enabled.

3. DEFAULT PRE-CONFIGURED TESTING CREDENTIALS

USER PERSONA	USERNAME	PASSWORD	DEFAULT VIEW	PRIMARY CAPABILITIES
State Administrator	`admin`	`admin123`	Admin Panel	Feed/Edit/Delete Jobs, Manage Students, Export CSV
Student (Candidate 1)	`priya_patel`	`student123`	Student Panel	View Dossier, 1-Click Apply, Track 4 Applications
Student (Candidate 2)	`rahul_sharma`	`student123`	Student Panel	Mechanical Eng Profile, View Recommended Feeds
Public Citizen	`None (Open)`	`None (Open)`	Public Portal	Browse OJAS Ticker, Search Jobs across 33 Districts
URL Query Switcher	`?view=admin`	`N/A`	Direct Routing	URL parameters allow direct bookmarking to any view

4. ARCHITECTURAL VERIFICATION & SIGN-OFF CHECKLIST

OFFICIAL ARCHITECTURAL SIGN-OFF

[PASS] Responsive Media Support: Tested across Desktop (>1200px), Tablet (991px), Mobile (768px), and Compact (480px).

[PASS] Admin Job Feed Engine: Full CRUD lifecycle verified; job vacancies immediately persist in LocalStorage.

[PASS] Add Student Option Removal: Compliance rule honored; students register exclusively via Public Portal.

[PASS] Student Dossier & 1-Click Apply: Candidate tracking, GTU verified credentials, and instantaneous state updates verified.

[PASS] State CSV Export Engine: Standard RFC 4180 compliant CSV generation tested with Gujarati candidate data.

[PASS] Zero External Dependencies: Platform operates 100% offline with zero CDN dependencies or backend servers.